

Lit Review

Paper 1: Characterization of I/O Behaviors in Cloud Storage Workloads

2023, IEEE Transactions on Computers

1. Read | Write | Mixed | Metadata Heavy

- **Covered in Paper?** Yes
- **Plain-English Definition:** Workloads are classified as read-dominated or write-dominated based on which operation type accounts for the majority of the total request count.
- **Exact Cutoff/Threshold:** Simple majority of request counts (e.g., Read-dominated if total reads > total writes).
- **Darshan Formula:** $\text{POSIX_BYTES_READS} > \text{POSIX_BYTES_WRITES}$ (for Read-dominated).
- **Location in Paper:** Section II-A; Table I.

2. ~~Small | Medium | Large I/O Size~~

- ~~~~Covered in Paper? No~~~~
- ~~~~Plain-English Definition:~~ Not analyzed in this paper; the authors focus on temporal arrival patterns rather than transaction sizes.~~~~~~
- ~~~~Exact Cutoff/Threshold:~~ Not analyzed in this paper.~~~~~~
- ~~~~Darshan Formula:~~ Not analyzed in this paper.~~~~~~
- ~~~~Location in Paper:~~ Not analyzed in this paper.~~~~~~

3. Operation Count Categories

- **Covered in Paper?** Yes
- **Plain-English Definition:** The paper analyzes the intensity of workloads by measuring the number of I/O requests per unit of time (second) to identify burstiness.
- **Exact Cutoff/Threshold:** Burstiness is identified when the number of requests per unit time fluctuates significantly across subintervals.
- **Darshan Formula:** $(\text{POSIX_READS} + \text{POSIX_WRITES}) / \text{Total_Seconds}$ (Average request rate).
- **Location in Paper:** Section III-A; Figure 5.

4. Time / Performance

- **Covered in Paper?** Yes
- **Plain-English Definition:** Performance is analyzed through inter-arrival times (the time between adjacent requests) and the Hurst parameter to identify self-similarity.
- **Exact Cutoff/Threshold:** Self-similarity is confirmed if the Hurst parameter (H) is in the range $(0.5, 1)$.
- **Darshan Formula:** $\text{POSIX_F_READ_START_TIMESTAMP}[i] - \text{POSIX_F_READ_START_TIMESTAMP}[i-1]$ (Inter-arrival time delta).
- **Location in Paper:** Section IV-C; Figure 8.

~~5. Sequential vs. Random Access~~

- ~~~~Covered in Paper? No~~~~
- ~~~~Plain-English Definition:~~ Not analyzed in this paper; while spatial patterns are mentioned as related work, this study focuses on temporal correlations.~~~~~~
- ~~~~Exact Cutoff/Threshold:~~ Not analyzed in this paper.~~~~~~
- ~~~~Darshan Formula:~~ Not analyzed in this paper.~~~~~~
- ~~~~Location in Paper:~~ Not analyzed in this paper.~~~~~~

~~6. Strided vs. Non-Strided Access~~

- ~~~~Covered in Paper? No~~~~
 - ~~~~Plain-English Definition:~~ Not analyzed in this paper.~~~~~~
 - ~~~~Exact Cutoff/Threshold:~~ Not analyzed in this paper.~~~~~~
 - ~~~~Darshan Formula:~~ Not analyzed in this paper.~~~~~~
 - ~~~~Location in Paper:~~ Not analyzed in this paper.~~~~~~
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Paper 2: An In-Depth Analysis of Cloud Block Storage Workloads in Large-Scale Production

2020, IEEE International Symposium on Workload Characterization (IISWC)

1. Read | Write | Mixed | Metadata Heavy

- **Covered in Paper?** Yes
- **Plain-English Definition:** Workloads are categorized by a "write-to-read ratio" comparing the total number of write operations to read operations.
- **Exact Cutoff/Threshold:** Write-dominant if the write-to-read ratio is > 1 ; very high write-dominance if ratio > 100 .

- **Darshan Formula:** $\text{POSIX_WRITES} / \text{POSIX_READS} > 1$.
- **Location in Paper:** Section III-C; Figure 4.

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2. Small | Medium | Large I/O Size

- **Covered in Paper?** Yes (add boundary values)
- **Plain-English Definition:** The paper defines "small-size" I/O based on individual request volumes falling below a specific kilobyte threshold.
- **Exact Cutoff/Threshold:** Small < 100 KiB; specifically, 75% of operations are ≤ 32 KiB for reads and ≤ 16 KiB for writes.
- **Darshan Formula:** $\text{POSIX_SIZE_READ_0_100} + \text{POSIX_SIZE_WRITE_0_100}$.
- **Location in Paper:** Section III-C; Figure 2.

3. Operation Count Categories

- **Covered in Paper?** Yes
- **Plain-English Definition:** Workload intensity is measured by the frequency of requests (req/s), comparing the maximum requests in a one-minute window to the overall average.
- **Exact Cutoff/Threshold:** High burstiness is identified if the "burstiness ratio" (peak intensity / average intensity) exceeds 100.
- **Darshan Formula:** $\text{Peak_Req_Per_Min} / ((\text{POSIX_READS} + \text{POSIX_WRITES}) / \text{Total_Seconds})$.
- **Location in Paper:** Section IV-A; Finding 2; Table II.

4. Time / Performance

- **Covered in Paper?** Yes
- **Plain-English Definition:** Performance is evaluated through "inter-arrival times," which is the elapsed time between two consecutive I/O requests for the same volume.
- **Exact Cutoff/Threshold:** Short-term burstiness is confirmed if the medians of the 25th, 50th, and 75th percentiles of inter-arrival times are lower than 1.3 ms.
- **Darshan Formula:** $\text{POSIX_F_READ_START_TIMESTAMP}[i] - \text{POSIX_F_READ_END_TIMESTAMP}[i-1]$ (Delta between adjacent operations).
- **Location in Paper:** Section IV-A; Finding 4; Figure 7.

5. Sequential vs. Random Access

- **Covered in Paper?** Yes
- **Plain-English Definition:** A request is classified as "random" if the distance between its offset and the offsets of the previous 32 requests exceeds a specific spatial threshold.
- **Exact Cutoff/Threshold:** Random if the minimum distance to the previous 32 offsets is > 128 KiB; "randomness ratio" is the percentage of such requests.
- **Darshan Formula:** $1 - (\text{POSIX_SEQ_READS} / \text{POSIX_READS})$ (Using inverse sequentiality as a proxy for the randomness ratio).
- **Location in Paper:** Section IV-B; Finding 8; Figure 10.

6. ~~Strided vs. Non-Strided Access~~

- ~~~~Covered in Paper? No~~~~
 - ~~~~Plain-English Definition: Not analyzed in this paper.~~~~
 - ~~~~Exact Cutoff/Threshold: Not analyzed in this paper.~~~~
 - ~~~~Darshan Formula: Not analyzed in this paper.~~~~
 - ~~~~Location in Paper: Not analyzed in this paper.~~~~
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Paper 3: Storage Workload Characterization and Consolidation in Virtualized Environments

2009, Workshop on Virtualization Performance: Analysis, Characterization, and Tools (VPACT)

1. Read | Write | Mixed | Metadata Heavy

- **Covered in Paper?** Yes
- **Plain-English Definition:** Workloads are classified by the numerical ratio of read requests to write requests observed at the hypervisor level.
- **Exact Cutoff/Threshold:** Example given for TPC-C data LUN where the ratio of reads to writes is exactly 2:1.
- **Darshan Formula:** $\text{POSIX_READS} / \text{POSIX_WRITES}$ (Read-to-Write Ratio).
- **Location in Paper:** Section 2; Section 3.3.

2. Small | Medium | Large I/O Size

- **Covered in Paper?** Yes
- **Plain-English Definition:** The authors use histogram bins to track individual request sizes, identifying specific application "page sizes" (like 2KB or 8KB).
- **Exact Cutoff/Threshold:** Common application sizes are 2KB and 8KB; sequential streams are coalesced into "larger IO requests (>256K)" by the storage system.

- **Darshan Formula:** `POSIX_SIZE_READ_0_100` (for small database IO) vs. `POSIX_SIZE_READ_100K_1M` (for coalesced 256K+ IO).
- **Location in Paper:** Section 2; Section 4.

3. Operation Count Categories

- **Covered in Paper?** Yes
- **Plain-English Definition:** The paper tracks the frequency of I/O arrivals (arrivals per second) and the number of outstanding (queued) operations at any given time.
- **Exact Cutoff/Threshold:** Arrivals per second are measured in 1-second and 30-second intervals; outstanding I/O (concurrency) is tracked in bins from 1 to 32+.
- **Darshan Formula:**
$$\frac{(\text{POSIX_READS} + \text{POSIX_WRITES})}{(\text{POSIX_F_CLOSE_END_TIMESTAMP} - \text{POSIX_F_OPEN_START_TIMESTAMP})}$$
 (Average operation frequency).
- **Location in Paper:** Section 2; Section 5.

4. Time / Performance

- **Covered in Paper?** Yes
- **Plain-English Definition:** Performance is defined by "IO latency" (time from VM issue to completion) and the "inter-arrival period" (time elapsed between subsequent requests).
- **Exact Cutoff/Threshold:** Inter-arrival times are measured in microseconds; latencies typically range from 0.5ms (logs) to 15ms (data reads).
- **Darshan Formula:**
$$\text{POSIX_F_READ_START_TIMESTAMP}[i] - \text{POSIX_F_READ_END_TIMESTAMP}[i-1]$$
 (Inter-arrival time logic).
- **Location in Paper:** Section 2; Section 3.1.

5. Sequential vs. Random Access

- **Covered in Paper?** Yes
- **Plain-English Definition:** Access is classified using "seek distance," which is the logical block distance between the current request and the very last I/O.
- **Exact Cutoff/Threshold:** "Small distances signify high locality" (sequential), while large distances in the histogram signify randomness.
- **Darshan Formula:**
$$\text{POSIX_SEQ_READS} / \text{POSIX_READS}$$
 (Sequentiality Ratio).
- **Location in Paper:** Section 2.

6. Strided vs. Non-Strided Access

- **Covered in Paper?** Yes
- **Plain-English Definition:** The authors identify stride patterns when I/O requests arrive at large but consistent offsets from previous requests.

- **Exact Cutoff/Threshold:** Microsoft Exchange exhibited a stride pattern where 20% of reads arrived within a consistent 32KB offset of the previous I/O.
 - **Darshan Formula:** Requires standard POSIX indicators (tracking repeated seek distance deltas in the histogram).
 - **Location in Paper:** Section 3.2.
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Paper 4: Extracting Flexible, Replayable Models from Large Block Traces

2012, FAST

1. Read | Write | Mixed | Metadata Heavy

- **Covered in Paper?** Yes
- **Plain-English Definition:** Workloads are characterized by the ratio of read operations to write operations and the statistical distribution of these request types.
- **Exact Cutoff/Threshold:** The authors calculate the probability of an operation being a read ($P(\text{read})$) based on total counts within a trace chunk.
- **Darshan Formula:** $\text{POSIX_READS} / (\text{POSIX_READS} + \text{POSIX_WRITES})$.
- **Location in Paper:** Section 1; Section 3 (Figure 1).

2. Small | Medium | Large I/O Size

- **Covered in Paper?** Yes
- **Plain-English Definition:** I/O size is treated as a core statistical distribution; the paper uses histograms to capture the frequency of specific request sizes.
- **Exact Cutoff/Threshold:** Example thresholds in the feature matrix use bins such as <4KB, 8KB, 12KB, and 16KB.
- **Darshan Formula:** Aggregation of specific buckets: $\text{POSIX_SIZE_READ_0_100}$ (for small) vs. $\text{POSIX_SIZE_READ_1M_4M}$ (for large).
- **Location in Paper:** Section 3; Figure 1.

3. Operation Count Categories

- **Covered in Paper?** Yes
- **Plain-English Definition:** The system explicitly tracks the count (frequency) of operations falling into specific size and offset categories within a multi-dimensional matrix.
- **Exact Cutoff/Threshold:** The frequency is calculated as the absolute number of requests per feature bin (e.g., "52 requests of size less than 4KB").

- **Darshan Formula:** `POSIX_READS` (Total Count) vs. `POSIX_BYTES_READ` (Total Volume).
- **Location in Paper:** Section 3; Figure 1.

4. Time / Performance

- **Covered in Paper?** Yes
- **Plain-English Definition:** Performance is modeled using inter-arrival times (the time elapsed between consecutive requests) to reproduce realistic system load and idling.
- **Exact Cutoff/Threshold:** Evaluation accuracy is measured by comparing throughput and latency, with a target error rate of less than 10-15%.
- **Darshan Formula:** `POSIX_F_READ_START_TIMESTAMP[i] - POSIX_F_READ_START_TIMESTAMP[i-1]` (Inter-arrival time).
- **Location in Paper:** Section 3; Section 5.

exhibits a highly sequential phase during its initialization startup

5. Sequential vs. Random Access

- **Covered in Paper?** Yes
- **Plain-English Definition:** Access patterns are classified using "seek distance" or "inter-arrival distance," defined as the logical offset difference between two consecutive requests.
- **Exact Cutoff/Threshold:** High sequentiality is identified when subsequent requests target adjacent or nearby blocks.
- **Darshan Formula:** `POSIX_SEQ_READS / POSIX_READS` (Sequentiality Ratio).
- **Location in Paper:** Section 3; Section 5.

~~6. Strided vs. Non-Strided Access~~

- ~~~~Covered in Paper? No~~~~
- ~~~~Plain-English Definition: Not analyzed in this paper.~~~~
- ~~~~Exact Cutoff/Threshold: Not analyzed in this paper.~~~~
- ~~~~Darshan Formula: Not analyzed in this paper.~~~~
- ~~~~Location in Paper: Not analyzed in this paper.~~~~

Paper 5: SSD-based Workload Characteristics and Their Performance Implications

2021, ACM Transactions on Storage (TOS)

1. Read | Write | Mixed | Metadata Heavy

- **Covered in Paper?** Yes
- **Plain-English Definition:** Workloads are characterized by their read/write ratio to assess device performance impacts like write amplification.
- **Exact Cutoff/Threshold:** While no single global cutoff is defined, the authors analyze specific workloads like RocksDB with an explicit split of 50% reads and 50% updates.
- **Darshan Formula:** $\text{POSIX_READS} / (\text{POSIX_READS} + \text{POSIX_WRITES})$.
- **Location in Paper:** Section 1; Section 2.2.

2. Small | Medium | Large I/O Size

- **Covered in Paper?** Yes
- **Plain-English Definition:** The paper evaluates request sizes relative to device page sizes (e.g., 4KB, 8KB, 16KB) to determine alignment costs.
- **Exact Cutoff/Threshold:** Median write sizes vary across repositories from 1 KB to 1,024 KB; 4 KB is frequently cited as the standard file-system page size.
- **Darshan Formula:** Sum of $\text{POSIX_SIZE_READ_0_100}$ through $\text{POSIX_SIZE_READ_1K_10K}$ (to capture 4KB-aligned "small" I/O).
- **Location in Paper:** Table 1; Section 4.

3. Operation Count Categories

- **Covered in Paper?** Yes
- **Plain-English Definition:** The analysis uses "virtual time," which is a counter incremented on each individual I/O request to measure logical locality.
- **Exact Cutoff/Threshold:** The study focuses on traces containing at least 10,000 total requests and 5,000 write requests.
- **Darshan Formula:** $\text{POSIX_READS} + \text{POSIX_WRITES}$ (Total request count used for virtual time windows).
- **Location in Paper:** Section 2.2; Section 5.1.

4. Time / Performance

- **Covered in Paper?** Yes
- **Plain-English Definition:** Performance is measured through inter-arrival times, sensing overhead ($\$t_R\$$), and serial access speed ($\$T_{\{\text{serial}\}}\$$) to compute flash read costs.
- **Exact Cutoff/Threshold:** Sensing overhead ($\$t_R\$$) for various flash technologies is typically 25–91 μsecs .
- **Darshan Formula:** $\max(\text{POSIX_F_CLOSE_END_TIMESTAMP}) - \min(\text{POSIX_F_OPEN_START_TIMESTAMP})$.
- **Location in Paper:** Section 1; Section 4.4.

5. Sequential vs. Random Access

- **Covered in Paper?** Yes
- **Plain-English Definition:** Sequentiality is the portion of sequential accesses in a workload, while "logical locality" characterizes the spatial relationship between random accesses.
- **Exact Cutoff/Threshold:** Logical locality is defined by the probability ($P_{\{D,T\}}$) that a page within distance D will be written within time window T .
- **Darshan Formula:** $\text{POSIX_SEQ_READS} / \text{POSIX_READS}$.
- **Location in Paper:** Section 1; Section 5.

6. Strided vs. Non-Strided Access

- **Covered in Paper?** No
 - **Plain-English Definition:** Not analyzed in this paper.
 - **Exact Cutoff/Threshold:** Not analyzed in this paper.
 - **Darshan Formula:** Not analyzed in this paper.
 - **Location in Paper:** Not analyzed in this paper.
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Paper 6: Extracting and characterizing I/O behavior of HPC workloads

2022, IEEE International Conference on Cluster Computing (CLUSTER)

1. Read | Write | Mixed | Metadata Heavy

- **Covered in Paper?** Yes
- **Plain-English Definition:** Workloads are classified by comparing the raw volume of bytes transferred (Read vs. Write) and the ratio of metadata operations to data operations.
- **Exact Cutoff/Threshold:** Metadata-heavy is identified when metadata operations dominate the time or count (e.g., CosmoFlow spends 98% of I/O time on metadata).
- **Darshan Formula:** $(\text{POSIX_OPENS} + \text{POSIX_F_CLOSE_START_TIMESTAMP}) / (\text{POSIX_READS} + \text{POSIX_WRITES} + \text{POSIX_OPENS} + \text{POSIX_F_CLOSE_START_TIMESTAMP})$.
- **Location in Paper:** Table III; Section IV-A-3.

2. Small | Medium | Large I/O Size

- **Covered in Paper?** Yes

- **Plain-English Definition:** Authors use histograms to bin I/O requests into sizes to identify if the workload is dominated by small transfers or large, high-bandwidth ones.
- **Exact Cutoff/Threshold:** Small is defined as $< 4\text{KB}$ or 64KB ; Large is defined as $> 16\text{MB}$.
- **Darshan Formula:** Sum of `POSIX_SIZE_READ_0_100` to `POSIX_SIZE_READ_1K_10K` (Small) vs. `POSIX_SIZE_READ_10M_100M` and up (Large).
- **Location in Paper:** Figure 1(a); Table VI.

3. Operation Count Categories

- **Covered in Paper?** Yes
- **Plain-English Definition:** The paper distinguishes between the volume of data moved and the absolute frequency of operations.
- **Exact Cutoff/Threshold:** Evaluated by "I/O ops distribution" (e.g., Montage MPI has 4M reads vs. 1M writes moving 21GB).
- **Darshan Formula:** `POSIX_READS` (Count) vs. `POSIX_BYTES_READ` (Volume).
- **Location in Paper:** Table III; Section V-B-1.

4. Time / Performance

- **Covered in Paper?** Yes
- **Plain-English Definition:** Performance is measured by the percentage of total job runtime consumed by I/O and the aggregate bandwidth achieved.
- **Exact Cutoff/Threshold:** I/O intensity is reported as "% of I/O time" (e.g., 11% for CM1, 75% for HACC).
- **Darshan Formula:** $\max(\text{POSIX_F_CLOSE_END_TIMESTAMP}) - \min(\text{POSIX_F_OPEN_START_TIMESTAMP})$ (Total I/O Window).
- **Location in Paper:** Table I; Section IV-A-1.

5. Sequential vs. Random Access

- **Covered in Paper?** Yes
- **Plain-English Definition:** Patterns are classified as sequential if the application accesses data in a continuous stream without significant seek operations.
- **Exact Cutoff/Threshold:** Most exemplar workloads are explicitly labeled as "Sequential I/O".
- **Darshan Formula:** `POSIX_SEQ_READS / POSIX_READS`.
- **Location in Paper:** Table I; Table VI.

6. Strided vs. Non-Strided Access

- **Covered in Paper?** Yes

- **Plain-English Definition:** The paper evaluates multidimensional data representations and discusses how storage "chunking" or "striping" optimizes these logical strides.
 - **Exact Cutoff/Threshold:** Defined by application data representation (e.g., 3D fixed arrays) and library-specific "chunking" configurations.
 - **Darshan Formula:** Requires standard POSIX histograms to detect repeating gaps, or specific MPI-IO module counter fields.
 - **Location in Paper:** Section I; Table VI.
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Paper 7: Workload Characterization for Enterprise Disk Drives

2018, ACM Transactions on Storage (TOS)

1. Read | Write | Mixed | Metadata Heavy

- **Covered in Paper?** Yes
- **Plain-English Definition:** Workloads are classified by the percentage of total commands or total bytes transferred that are reads versus writes.
- **Exact Cutoff/Threshold:** Reads are the dominant workload, accounting for 60%–80% of total bytes transferred and approximately 80% of all drive accesses.
- **Darshan Formula:** $\text{POSIX_READS} / (\text{POSIX_READS} + \text{POSIX_WRITES}) \geq 0.8$ (for access dominance) or $\text{POSIX_BYTES_READ} / (\text{POSIX_BYTES_READ} + \text{POSIX_BYTES_WRITTEN}) \geq 0.6$ (for volume dominance).
- **Location in Paper:** Section 2.3.

2. Small | Medium | Large I/O Size

- **Covered in Paper?** Yes
- **Plain-English Definition:** The author distinguishes between "small runs" (individual command sizes) and "long runs" (aggregated transfer sizes).
- **Exact Cutoff/Threshold:** Small access < 100KB; Long/Large run > 100KB.
- **Darshan Formula:** Small: $\text{Sum of POSIX_SIZE_READ_0_100}$ | Large: $\text{Sum of POSIX_SIZE_READ_100K_1M to POSIX_SIZE_READ_1G_PLUS}$.
- **Location in Paper:** Section 4.

3. Operation Count Categories

- **Covered in Paper?** Yes
- **Plain-English Definition:** The paper explicitly compares the frequency of operation counts against the volume of data moved.

- **Exact Cutoff/Threshold:** 90% of operations (runs) are < 100KB, yet 75%–95% of total bytes are transferred in the remaining operations that are > 100KB.
- **Darshan Formula:** $(\text{Sum of POSIX_SIZE_READ_100K_1M} \dots) / \text{POSIX_READS} \leq 0.1$ (Frequency) vs. $(\text{Bytes in those buckets}) / \text{POSIX_BYTES_READ} \geq 0.75$ (Volume).
- **Location in Paper:** Section 4.

4. Time / Performance

- **Covered in Paper?** Yes
- **Plain-English Definition:** Performance is measured as "Throughput" (bytes per unit time) and "IOPS" (commands per second) averaged over fixed 2-hour intervals.
- **Exact Cutoff/Threshold:** Throughput = Total bytes / Recording Interval (7,200s); IOPS = Total commands / (7,200s - Idle time).
- **Darshan Formula:** $(\text{POSIX_BYTES_READ} + \text{POSIX_BYTES_WRITTEN}) / (\text{POSIX_F_CLOSE_END_TIMESTAMP} - \text{POSIX_F_OPEN_START_TIMESTAMP})$.
- **Location in Paper:** Section 2.3.

5. Sequential vs. Random Access

- **Covered in Paper?** Yes
- **Plain-English Definition:** A command is "random" if it requires a seek; it is "sequential" if it starts exactly where the previous transfer ended.
- **Exact Cutoff/Threshold:** Random = Seek required; Sequential = Gap of 0 blocks; Near-Sequential = Gap of a few blocks.
- **Darshan Formula:** $\text{POSIX_SEQ_READS} / \text{POSIX_READS}$ (Sequentiality Ratio).
- **Location in Paper:** Section 2.3.

~~6. Strided vs. Non-Strided Access~~

- ~~~~Covered in Paper? No~~~~
- ~~~~Plain-English Definition:~~ Not analyzed in this paper; the author mentions "near-sequential" but does not evaluate formal strided HPC patterns.~~
- ~~~~Exact Cutoff/Threshold:~~ Not analyzed in this paper.~~
- ~~~~Darshan Formula:~~ Not analyzed in this paper.~~
- ~~~~Location in Paper:~~ Not analyzed in this paper.~~

Paper 8: Optimization of Reading Data via Classified Block Access Patterns in File Systems

1. Read | Write | Mixed | Metadata Heavy

- **Covered in Paper?** Yes
- **Plain-English Definition:** Workloads are classified by a "Read Ratio," representing the percentage of total I/O requests that are read operations.
- **Exact Cutoff/Threshold:** The study specifically selects "read-intensive workloads" with read ratios ranging from 64% to 99%.
- **Darshan Formula:** $\text{POSIX_READS} / (\text{POSIX_READS} + \text{POSIX_WRITES})$.
- **Location in Paper:** Table 1; Section III.

2. Small | Medium | Large I/O Size

- **Covered in Paper?** Yes
- **Plain-English Definition:** The paper tracks the "Size per Read" for various traces to evaluate how transaction volume affects performance.
- **Exact Cutoff/Threshold:** Average read sizes are reported for traces, ranging from "small" 8.2 KB (TPC-C) to 59.3 KB (src1).
- **Darshan Formula:** $\text{POSIX_BYTES_READ} / \text{POSIX_READS}$ (Average read size).
- **Location in Paper:** Table 1.

3. Operation Count Categories

- **Covered in Paper?** Yes
- **Plain-English Definition:** The authors analyze the frequency of block access events within a "prediction window" to identify correlations.
- **Exact Cutoff/Threshold:** Access patterns are constrained to a specific number of access events (N), specifically within the interval of.
- **Darshan Formula:** $(\text{POSIX_READS} + \text{POSIX_WRITES})$ (Total count used to define the access series).
- **Location in Paper:** Section III; Section II-B.

4. Time / Performance

- **Covered in Paper?** Yes
- **Plain-English Definition:** Performance is measured by the "average I/O response time".
- **Exact Cutoff/Threshold:** The mechanism targets a reduction in average response time (e.g., achieving 14.6%–17.9% reduction compared to sequential prefetching).
- **Darshan Formula:** $(\text{POSIX_F_READ_END_TIMESTAMP} - \text{POSIX_F_READ_START_TIMESTAMP})$ (Summed and averaged).
- **Location in Paper:** Table 2; Figure 4.

5. Sequential vs. Random Access

- **Covered in Paper?** Yes
- **Plain-English Definition:** Access is classified as sequential if it targets consecutive blocks; the paper targets non-sequential correlations that sequential prefetching misses.
- **Exact Cutoff/Threshold:** Sequential prefetching is defined as reading consecutive blocks after a cache miss.
- **Darshan Formula:** $\text{POSIX_SEQ_READS} / \text{POSIX_READS}$.
- **Location in Paper:** Section III.

6. Strided vs. Non-Strided Access

- **Covered in Paper?** Yes
 - **Plain-English Definition:** The paper acknowledges "stride read requests" as a specific non-sequential pattern where rules remain unchanged despite gaps.
 - **Exact Cutoff/Threshold:** Strided requests are identified qualitatively as a primary reason for the failure of standard sequential prefetching.
 - **Darshan Formula:** Requires standard POSIX histograms to detect recurring seek distances (gaps) between requests.
 - **Location in Paper:** Section III.
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Paper 9: Characterization of storage workload traces from production Windows Servers

2008, IEEE International Symposium on Workload Characterization

1. Read | Write | Mixed | Metadata Heavy

- **Covered in Paper?** Yes
- **Plain-English Definition:** Workloads are classified by the ratio of read requests to write requests and metadata update volume.
- **Exact Cutoff/Threshold:** Metadata-heavy is identified when system-level metadata updates (e.g., NTFS logs) significantly skew expected ratios (e.g., from 1:1 to 1:2).
- **Darshan Formula:** $\text{POSIX_READS} / (\text{POSIX_READS} + \text{POSIX_WRITES})$.
- **Location in Paper:** Section IV; Table I.

2. Small | Medium | Large I/O Size

- **Covered in Paper?** Yes
- **Plain-English Definition:** The paper tracks request size distributions to identify common sizes dictated by application logic or file system cluster sizes.

- **Exact Cutoff/Threshold:** Common modes include 4 KB (NTFS default), 8 KB (Exchange), and 64 KB (Large transfer mode).
- **Darshan Formula:** $\text{POSIX_SIZE_READ_0_100}$ (Aggregation of common mode buckets).
- **Location in Paper:** Table I; Section V-A.

3. Operation Count Categories

- **Covered in Paper?** Yes
- **Plain-English Definition:** The paper distinguishes between "Total IOs" (frequency) and "Total GB Transferred" (volume).
- **Exact Cutoff/Threshold:** Assessed by comparing "Total GB Transferred" against "Total IOs" to find average request intensity.
- **Darshan Formula:** $\text{POSIX_BYTES_READ} / \text{POSIX_READS}$.
- **Location in Paper:** Table I.

4. Time / Performance

- **Covered in Paper?** Yes
- **Plain-English Definition:** Performance is measured via average response times and the Hurst parameter to assess temporal self-similarity.
- **Exact Cutoff/Threshold:** Strong self-similarity (burstiness) is confirmed if the calculated Hurst parameter (H) is significantly greater than 0.5.
- **Darshan Formula:** $\text{POSIX_F_READ_START_TIMESTAMP}[i] - \text{POSIX_F_READ_START_TIMESTAMP}[i-1]$ (Inter-arrival time).
- **Location in Paper:** Table I; Section VI-A.

5. Sequential vs. Random Access

- **Covered in Paper?** Yes
- **Plain-English Definition:** Access is classified as sequential if the logical block offset between consecutive requests is exactly zero.
- **Exact Cutoff/Threshold:** Measured as the "% sequential IOs initiated" (e.g., 4%–32% depending on the specific workload).
- **Darshan Formula:** $\text{POSIX_SEQ_READS} / \text{POSIX_READS}$.
- **Location in Paper:** Table I; Section V-A.

6. Strided vs. Non-Strided Access

- **Covered in Paper?** Yes
- **Plain-English Definition:** The paper discusses detecting "fragmented" or "mixed" streams by comparing request offsets.
- **Exact Cutoff/Threshold:** Identified qualitatively as "jumps" between requests, often occurring in 4 KB multiples.

- **Darshan Formula:** Requires monitoring recurring seek distances (jumps) in the POSIX access histograms.
- **Location in Paper:** Section IV; Section V-A.

Paper 10: Understanding and Predicting Cross-Application I/O Interference in HPC Storage Systems

2008, IEEE International Symposium on Workload Characterization

1. Read | Write | Mixed | Metadata Heavy

- **Covered in Paper?** Yes.
- **Plain-English Definition:** Workloads are classified by the individual and combined **counts of different request types**, specifically distinguishing between **data operations** (reads and writes) and **metadata operations** (such as opens, closes, and stats).
- **Exact Cutoff/Threshold:** The study utilizes IO500 benchmark tasks like "ior-easy-write" (data-intensive) and "mdt-easy" (metadata-intensive) to simulate and categorize specific interference types.
- **Darshan Formula:** $POSIX_OPENS + POSIX_STATS + POSIX_F_CLOSE_START_TIMESTAMP$ (to calculate metadata intensity) versus $POSIX_READS + POSIX_WRITES$ (for data intensity).
- **Location in Paper:** Section III-A; Figure 1.

2. Small | Medium | Large I/O Size

- **Covered in Paper?** Yes.
- **Plain-English Definition:** I/O size is characterized by the **sum of bytes** conducted by read and write requests within a specific periodic time window.
- **Exact Cutoff/Threshold:** The framework relies on benchmark configurations where "ior-easy" represents large, sequential transfers and "ior-hard" represents small, unaligned, or "harder" transaction patterns.
- **Darshan Formula:** $POSIX_BYTES_READ / POSIX_READS$ (to determine average read size).
- **Location in Paper:** Section III-A; Section IV-2.

3. Operation Count Categories

- **Covered in Paper?** Yes.
- **Plain-English Definition:** This category tracks the **absolute frequency of I/O requests** (read, write, and metadata) aggregated over a user-defined time interval to contextualize workload intensity.

- **Exact Cutoff/Threshold:** Performance impacts are grouped into categories based on **slowdown ratios**: mild ($\leq 2x$), moderate ($2x-5x$), and severe ($\geq 5x$).
- **Darshan Formula:** $POSIX_READS + POSIX_WRITES$ (Total operations per time window).
- **Location in Paper:** Section III-A; Section IV-B.

4. Time / Performance

- **Covered in Paper?** Yes.
- **Plain-English Definition:** Performance is defined by the **actual I/O time** (duration of individual requests), overall throughput, and IOPS recorded within the monitoring window.
- **Exact Cutoff/Threshold:** The **degradation level** is calculated as the average ratio of I/O time during interference compared to a baseline execution time ($Level_{\{degrade\}} = Avg(iotime_{\{interference\}} / iotime_{\{base\}})$).
- **Darshan Formula:** $(POSIX_F_READ_END_TIMESTAMP - POSIX_F_READ_START_TIMESTAMP)$ for calculating the duration of individual operations.
- **Location in Paper:** Figure 1; Section III-D.

5. Sequential vs. Random Access

- **Covered in Paper?** Yes.
- **Plain-English Definition:** Access patterns are broadly categorized as **sequential** (easy) or **random/unaligned** (hard) based on their spatial behavior during benchmark tasks.
- **Exact Cutoff/Threshold:** Sequential behavior is defined by the **"ior-easy"** pattern; random access is simulated using the **"ior-hard"** task set.
- **Darshan Formula:** $POSIX_SEQ_READS / POSIX_READS$ (Sequentiality Ratio).
- **Location in Paper:** Section IV-2; Section V.

6. Strided vs. Non-Strided Access

- **Covered in Paper?** Yes.
- **Plain-English Definition:** The paper identifies **strided or "hard" access patterns** through specific IO500 tasks that induce contention on metadata or object storage targets via non-contiguous patterns.
- **Exact Cutoff/Threshold:** Qualitatively defined by the **IO500 "hard" task configurations**, which are used to measure extreme performance variance (e.g., slowdowns up to 40.9x).
- **Darshan Formula:** Requires fine-grained analysis of **Darshan DXT logs** to track individual offset jumps between operations.
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